By Prop. I and Cor. 4 the centripetal force in a least arc is as the versed sine of that arc taken in equal times. By Lemma VII the versed sine is as the arc squared divided by the diameter, so F answers to $\mathbf{arc}^2/D$. Since the arc is as $v\,t$ and the diameter is as the radius, the times cancel and the diameters give the r, leaving the v^2 standing as the heart. The proof chain therefore closes on $F \propto v^2/r$.